



Gaming Addiction Prediction Using Random Forest Classifier

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Warm regards,

Pranish Bhuju

Abstract

This report investigates the prediction of gaming addiction severity using a Random Forest Classifier applied to a structured behavioural dataset of 250 online gaming users. Once duplicate entries and null values were dropped 141 clean records remained for analysis. There are 48 attributes that comprise gaming attributes, psychographic factors, behavior related characteristics, and demographic attributes, which assist in the classification of the extent of their addiction into three categories, such as Mild, Moderate, and Severe.

Following systematic data pre-processing including duplicate removal, missing value handling, and label encoding of categorical variables the Random Forest model was trained on 112 records and evaluated on the remaining 29. The classifier consisted of 500 decision trees with a maximum depth of 12 and balanced class weighting to address the severe class imbalance where Severe contained only 3 records.

The model demonstrated strong performance, correctly identifying 26 out of 29 test cases to reach an accuracy of 89.66% and a mean confidence level of 78.12%. Its most significant predictive variables the dopamine dependency index, peak consecutive gaming hours, and daily playtime duration align with established research on gaming disorder. Nevertheless, one of the main weaknesses of the model was the inability to accurately predict "Severe" classifications due to a limited sample size, which remains an important area of future research.

1. Domain Description

The domain of this study can be described as a point where behaviour sciences, digital health, and machine learning applications overlap. There have been rapid developments in online gaming over the last two decades, such that the total number of gamers across the globe has already surpassed three billion (Newzoo, 2023). Despite the various advantages that can be associated with playing video games, some people become prone to problematic gaming behaviours.

World Health Organization officially recognized the term "Gaming Disorder" as a disease under the 11th revision of the International Classification of Diseases (ICD-11) in 2019 as an issue characterized by loss of control over gaming behaviors, preference towards gaming over other activities, and continued use of gaming regardless of negative consequences (WHO, 2019). The clinical recognition of the condition has resulted in a significant amount of academic research on the behaviors and psychological traits that set apart recreational gaming from addiction. This study uses the Random Forest Classifier algorithm due to its suitability with multidimensional datasets.

2. Problem Definition

The rapid growth of online gaming has raised pressing concerns arising in relation to the possible effects that could be exerted on the users psychological well being, efficiency at work, and general way of life. Though gaming might be considered an enjoyable and entertaining hobby, it can cause behavioral addiction, insomnia, stress, and diminished productivity.

The study aims to answer the following questions:

- What behavioral attributes make the most substantial contribution to gaming addiction?
- What is the correlation between gaming addiction and various lifestyle aspects including sleeping, stressing, and productivity?
- Is it possible to use machine learning techniques to predict levels of addiction?

3. Literature review

3.1 Gaming Disorder and Behavioural Research

There have been increasing numbers of studies done on gaming addiction since the year 2000. The groundbreaking work carried out by Kuss & Griffiths (2012) showed that the disorder is linked to mental illnesses such as anxiety and depression. Following this, Lemmens et al. (2009) created an addiction scale which helped establish the basis on how the quantifiable construct is measured.

Recent works for example those done by Andreassen et al. (2016), have shown that impulse control and lack of self-control play key roles in addiction. This helps explain why the two are key constructs used in this study.

3.2 Random Forest in Gaming Addiction Classification

Random Forest, invented by Breiman (2001), leverages the use of decision trees in an ensemble fashion to improve predictive performance and reduce issues of overfitting using bagging and subspace sampling. As noted by Liaw and Wiener (2002), one of the benefits of the Random Forest model is its intrinsic feature importance metrics, especially valuable when applied to behavioral research to extract meaningful risk factors.

With respect to online gaming addiction, the work by Ding et al. (2020) showed that Random Forest is capable of classifying online gaming addiction with more than 82% precision and stressed the importance of predictors such as screen usage and emotional regulation. It corroborates results from the work by King and Delfabbro (2014),

where it was confirmed that session length and night time gaming were valid predictors.

4. Dataset Description

The data set is Gaming Addiction Data Set and contains information on 250 records with data on online game addiction, psychological well being lifestyle factors, and real world performance. There were 141 records left after cleaning the data by removing the duplicate entries and missing observations. There are 48 attributes in the total data set however only those attributes that are relevant to this study have been included excluding the identifier attribute and the addiction labels. The attributes used in this study are:

Column	Type	Description	Sample Values
age	Integer	Age of the user in years	13 – 40
gender	Categorical	Gender identity of the user	Male, Female, Non-binary, Prefer not to say
daily_playtime_hours	Float	Average hours of gaming per day	0.5 – 11.9
weekly_play_sessions	Integer	Number of gaming sessions per week	1 – 15
late_night_sessions_hours	Float	Hours gaming during late-night periods	0.0 – 5.2
stress_score	Float	Self-reported stress level (1–10 scale)	1.0 – 10.0
dopamine_dependency_index	Float	Reward-seeking and craving intensity	1.0 – 8.6
self_control_score	Float	Ability to regulate impulses (1–10 scale)	1.0 – 10.0

impulsiveness_score	Float	Tendency towards impulsive behaviour	1.0 – 10.0
---------------------	-------	--------------------------------------	------------

Column	Type	Description	Sample Values
anxiety_level	Float	Self-reported anxiety (1–10 scale)	1.0 – 10.0
depression_indicator	Float	Depressive symptom severity score	1.0 – 8.6
sleep_hours	Float	Average nightly sleep duration in hours	3.4 – 9.5
productivity_drop_percent	Float	Estimated drop in work/academic productivity	0.0 – 44.2
screen_time_total_hours	Float	Total daily screen time across all devices	1.2 – 15.1
burnout_probability	Float	Predicted likelihood of burnout (0–1)	0.32 – 1.0
mental_health_risk_score	Float	Composite mental health risk score	0.12 – 0.92
addiction_severity	Categorical	TARGET – addiction severity classification	Mild, Moderate, Severe

Table 1: Columns used in this analysis - name,type,description and sample values

5. Data Pre-Processing

5.1 Library Imports and Setup

The analysis was conducted using Python. The following libraries were imported to support data manipulation, visualisation, and machine learning:

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder

from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import (
    accuracy_score,
    classification_report,
    confusion_matrix
)
```

Figure 1 : Library imports and setup

5.2 Loading the Dataset

The dataset was loaded using the Pandas `read_csv()` function. An initial inspection of the first five rows (`df.head()`) was performed to confirm the structure and verify column names.

```
df = pd.read_csv("gaming_addiction.csv")

print("Dataset Shape:", df.shape)

df.head()

Dataset Shape: (250, 49)
```

	user_id	age	gender	country	occupation	income_level	years_gaming	preferred_genre	platform	device_type	...	absenteeism_days	internet_speed_mbps
0	USR000001	21	Male	India	Employed	Middle	9	Sandbox	PC	Laptop	...	7	39.9
1	USR000002	25	Male	India	Employed	Lower-Middle	13	RPG	Mobile	Mixed	...	6	71.5
2	USR000003	26	Male	Brazil	Employed	Middle	14	RPG	PC+Mobile	High-end PC	...	7	119.4
3	USR000004	22	Male	South Korea	Employed	Upper-Middle	10	Strategy	PC+Mobile	Mobile	...	8	136.5
4	USR000005	17	Female	India	Student	Middle	5	Strategy	PC	Laptop	...	6	78.8

screen_time_total_hours	behavioral_cluster	addiction_score	addiction_binary	addiction_severity	burnout_probability	mental_health_risk_score	churn_probability
5.4	Casual Enjoyer	27.61	0	Mild	1.0	0.920	1.000
13.4	Streamer/Creator	55.51	1	Moderate	1.0	0.515	0.813
12.3	Streamer/Creator	45.85	0	Moderate	1.0	0.720	0.947
6.9	Toxic Competitor	39.87	0	Mild	1.0	0.520	0.660
9.3	Competitive Grinder	46.97	0	Moderate	1.0	0.585	0.867

Figure 2 : First five rows of Gaming Addiction Dataset

5.3 Missing Value Detection

A missing value audit was conducted to identify any null entries across the dataset. Both a summary count and a heatmap visualisation were used to inspect the data for completeness.

```
missing = df.isnull().sum()

print(missing[missing > 0])
print("Total Missing Value:",df.isnull().sum().sum())
```

```
plt.figure(figsize=(12,5))

sns.heatmap(
    df.isnull(),
    cbar=False
)

plt.title("Missing Values")
plt.show()
```

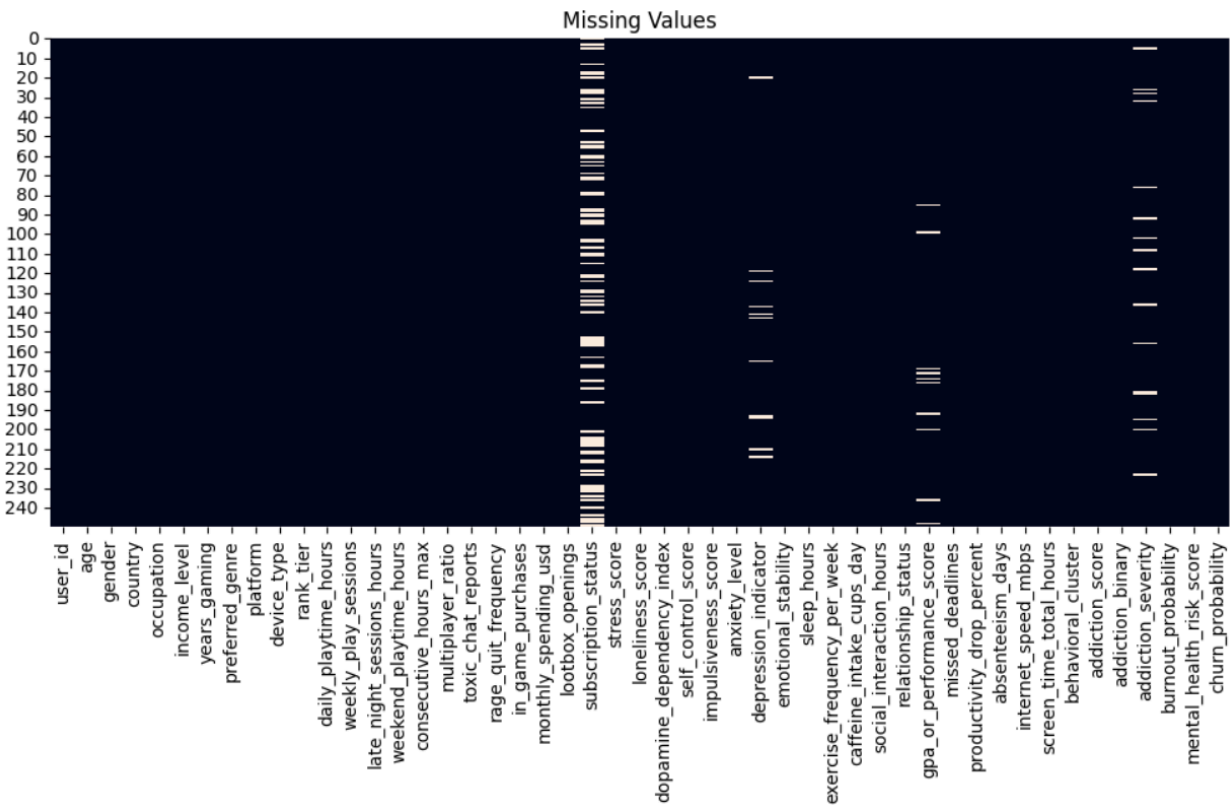


Figure 3 : Missing Values Heatmap

5.4 Duplicate Record Removal

The dataset was inspected for fully duplicated rows. Any such records were removed to ensure that the training and testing data reflected unique user observations. Missing values were also dropped at this stage to produce a clean working dataset.

```
# Remove completely identical rows (Duplicates)
df = df.drop_duplicates()

# Remove rows with empty/blank cells (Missing Values)
df = df.dropna()

# Check the final clean shape
print("Remaining missing values:", df.isnull().sum().sum())
print("Final Cleaned Dataset Shape:", df.shape)

Remaining missing values: 0
Final Cleaned Dataset Shape: (141, 49)
```

Figure 4 : Output after duplicate and missing value removal

5.5 Descriptive Statistics

Summary statistics were generated for all numeric attributes through the use of `df.describe()`, with the results arranged transversely for convenience of viewing. This helped gain a first impression of the nature of each attribute.

	count	mean	std	min	25%	50%	75%	max
age	141.0	22.163121	5.666473	13.000	18.000	22.000	27.000	40.00
years_gaming	141.0	9.624113	5.658797	1.000	6.000	9.000	15.000	25.00
daily_playtime_hours	141.0	6.469504	2.420624	0.500	4.900	6.700	8.200	11.80
weekly_play_sessions	141.0	7.503546	3.381935	1.000	5.000	7.000	10.000	15.00
late_night_sessions_hours	141.0	2.238298	1.110962	0.000	1.400	2.100	3.000	5.20
weekend_playtime_hours	141.0	8.596454	3.723294	0.500	6.000	8.400	10.600	20.00
consecutive_hours_max	141.0	10.318440	4.387215	1.000	7.100	10.800	13.400	19.40
multiplayer_ratio	141.0	0.469362	0.221773	0.000	0.320	0.460	0.630	1.00
toxic_chat_reports	141.0	3.297872	5.714324	0.000	0.000	1.000	3.000	32.00
rage_quit_frequency	141.0	2.134752	2.397081	0.000	0.000	1.000	3.000	10.00
in_game_purchases	141.0	4.141844	4.622278	0.000	0.000	3.000	7.000	19.00
monthly_spending_usd	141.0	41.803546	34.030209	0.000	15.100	36.450	58.740	170.51
lootbox_openings	141.0	8.666667	7.956728	0.000	2.000	7.000	13.000	37.00
stress_score	141.0	6.814184	1.653549	2.000	5.700	7.000	7.900	10.00
loneliness_score	141.0	5.077305	2.029651	1.000	3.800	5.000	6.600	10.00
dopamine_dependency_index	141.0	4.396454	1.618880	1.000	3.200	4.400	5.400	8.60
self_control_score	141.0	6.460284	1.828422	1.100	5.100	6.600	7.800	10.00
impulsiveness_score	141.0	5.009220	1.756087	1.000	3.800	5.100	6.400	9.40
anxiety_level	141.0	5.569504	1.698064	1.300	4.400	5.500	6.500	10.00
depression_indicator	141.0	4.599291	1.697035	1.000	3.500	4.500	5.900	8.60
emotional_stability	141.0	6.485816	1.757376	3.000	5.100	6.700	7.700	10.00
sleep_hours	141.0	6.735461	1.156234	3.400	5.900	6.800	7.600	9.20
exercise_frequency_per_week	141.0	1.255319	1.250167	0.000	0.000	1.000	2.000	5.00
caffeine_intake_cups_day	141.0	2.460993	1.662690	0.000	1.000	2.000	4.000	7.00
social_interaction_hours	141.0	2.214894	1.534644	0.000	1.000	2.000	3.500	6.60
gpa_or_performance_score	141.0	3.724823	0.487820	2.250	3.440	4.000	4.000	4.00
missed_deadlines	141.0	3.063830	2.043152	0.000	1.000	3.000	4.000	8.00
productivity_drop_percent	141.0	16.529078	10.427646	0.000	8.500	16.700	24.600	41.70
absenteeism_days	141.0	5.567376	3.078276	0.000	4.000	6.000	8.000	14.00
internet_speed_mbps	141.0	103.161702	57.174868	5.000	67.100	102.300	137.000	255.50
screen_time_total_hours	141.0	8.558156	2.653492	2.700	6.700	8.300	10.800	13.90
addiction_score	141.0	39.941915	10.120066	20.200	31.340	40.240	45.880	68.00
addiction_binary	141.0	0.148936	0.357295	0.000	0.000	0.000	0.000	1.00
burnout_probability	141.0	0.999617	0.004548	0.946	1.000	1.000	1.000	1.00
mental_health_risk_score	141.0	0.508440	0.134216	0.120	0.405	0.505	0.600	0.87
churn_probability	141.0	0.800773	0.112092	0.472	0.743	0.814	0.879	1.00

Figure 5 : Descriptive Statistics Output (`df.describe().T`) – mean, std, min, max for all numeric features

5.6 Label Encoding of Categorical Variables

The dataset contains several categorical string column including gender, country, occupation, preferred_genre, platform, and addiction_severity. These were transformed into numeric representations using Scikit Learn's LabelEncoder and applied to every column with object dtype. This encoding step was essential prior to correlation analysis and model training.

```
le = LabelEncoder()

for col in df.columns:
    if df[col].dtype == "object":
        df[col] = le.fit_transform(df[col].astype(str))
```

6. Descriptive Analysis Techniques

6.1 Exploratory Data Analysis (EDA)

6.1.1 Addiction Severity Distribution

A count plot of the target variable addiction_severity was generated to understand the class distribution across the dataset. The visualisation confirmed a heavily imbalanced spread: Mild accounted for 70 records, Moderate for 68, and Severe for only 3.

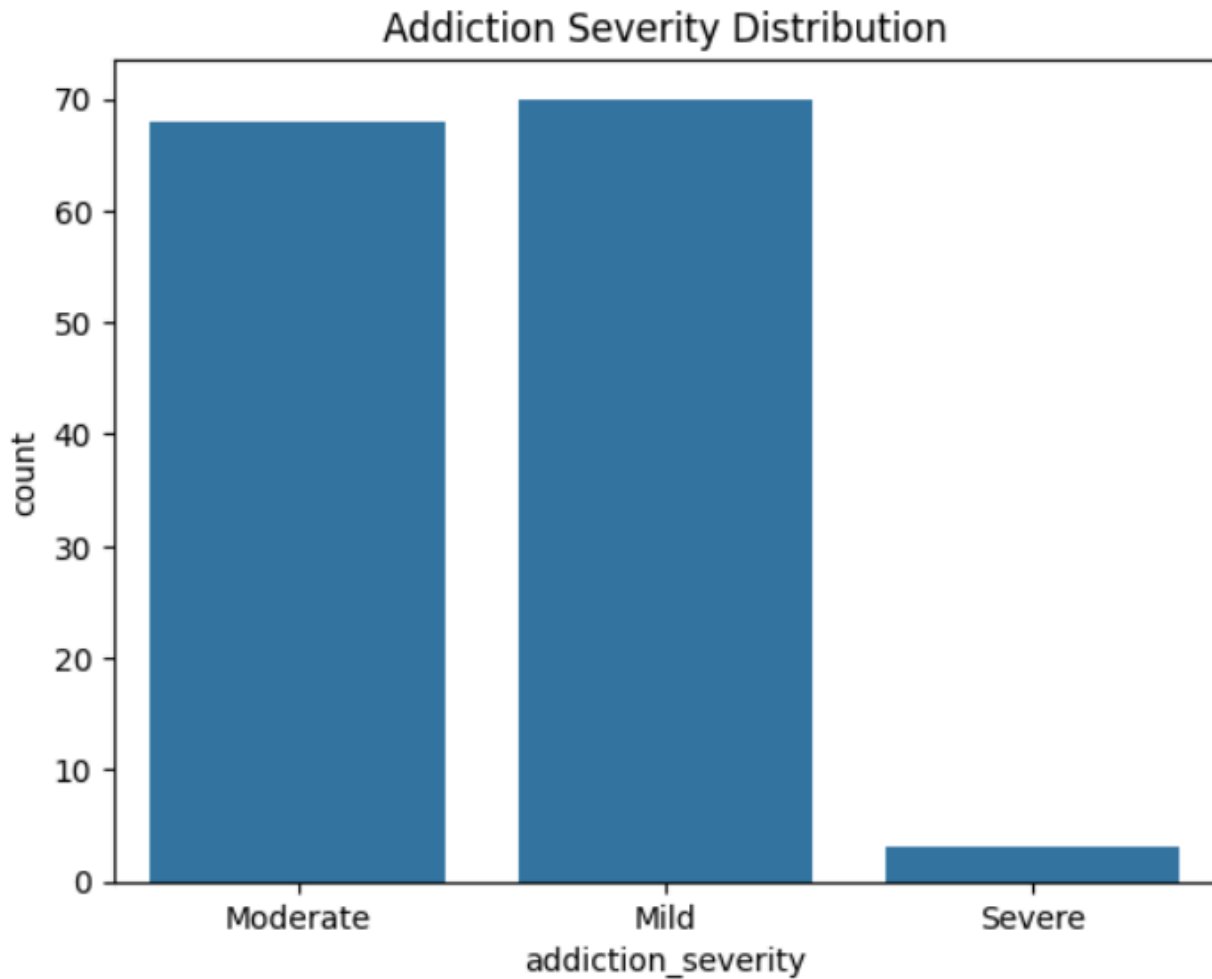


Figure 6 : Addiction Severity Distribution – Mild, Moderate, and Severe class counts

6.1.2 Age Distribution

According to the age histogram, it was noted that there existed a distribution with a positive skew, where the mode of the distribution falls between 20 years and 28 years.

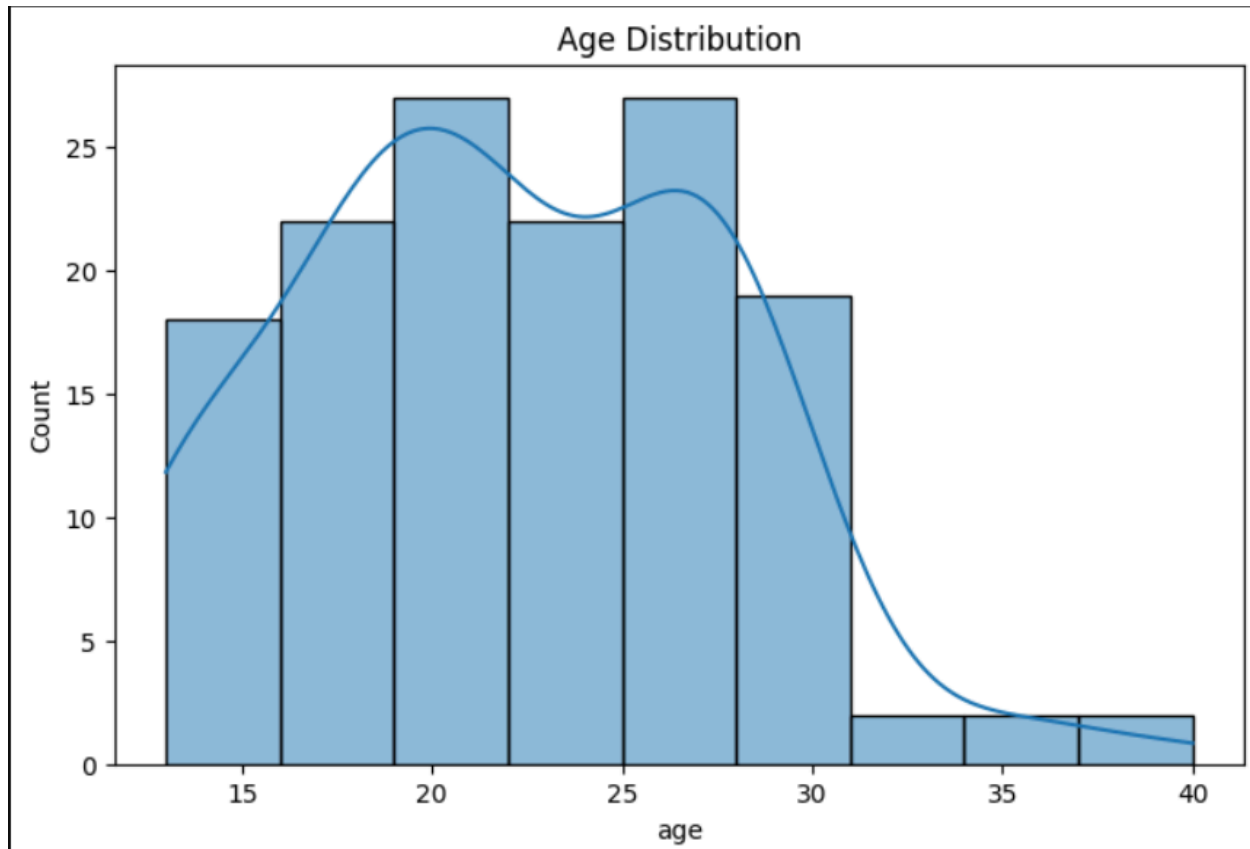


Figure 7 : Age Distribution Histogram

6.1.3 Daily Playtime Distribution

From the daily playtime distribution histogram it can be seen that there is wide variation, with some days having less than 2 hours and some others more than 14 hours daily.

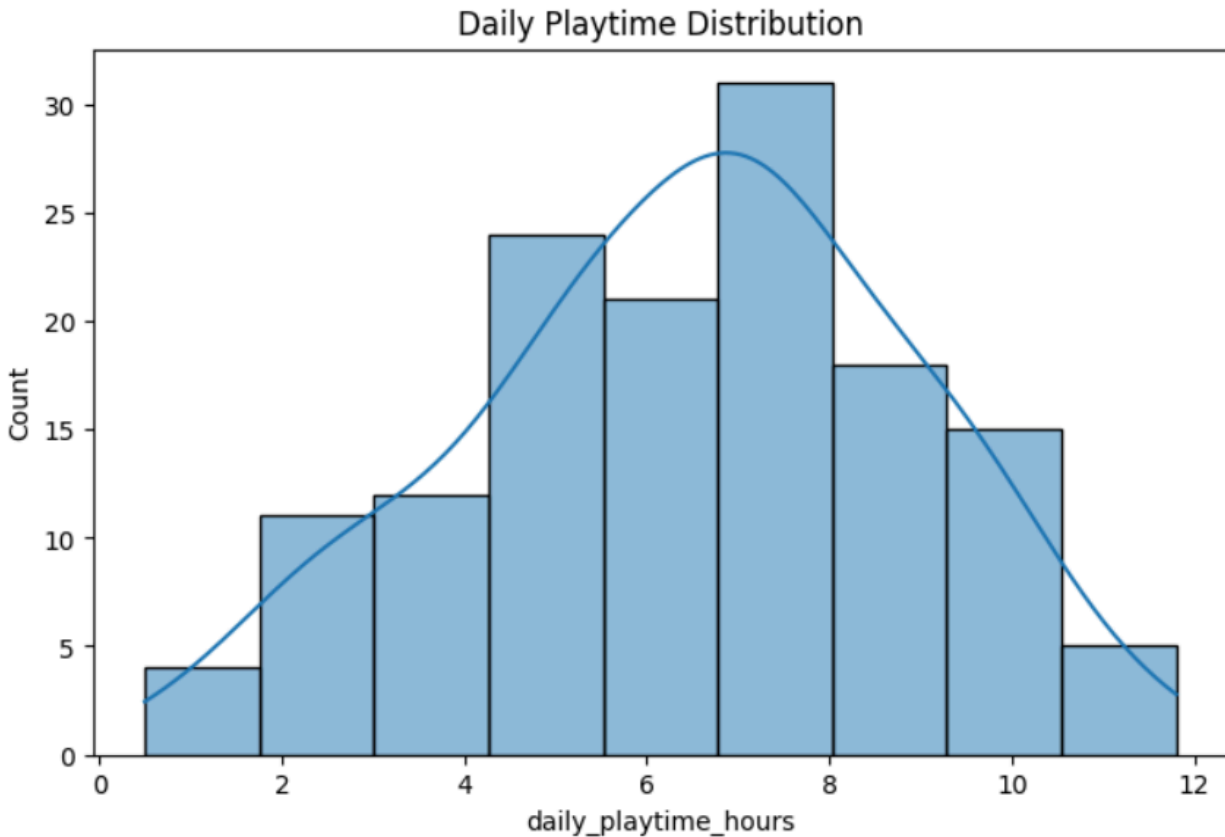


Figure 8 : Daily Playtime Distribution Histogram

6.1.4 Sleep Hours Distributions

Data from the sleep hours histogram reveals that many individuals average less than six hours of sleep daily, a figure notably lower than the 7–9 hours recommended by the National Sleep Foundation (Hirshkowitz et al., 2015).

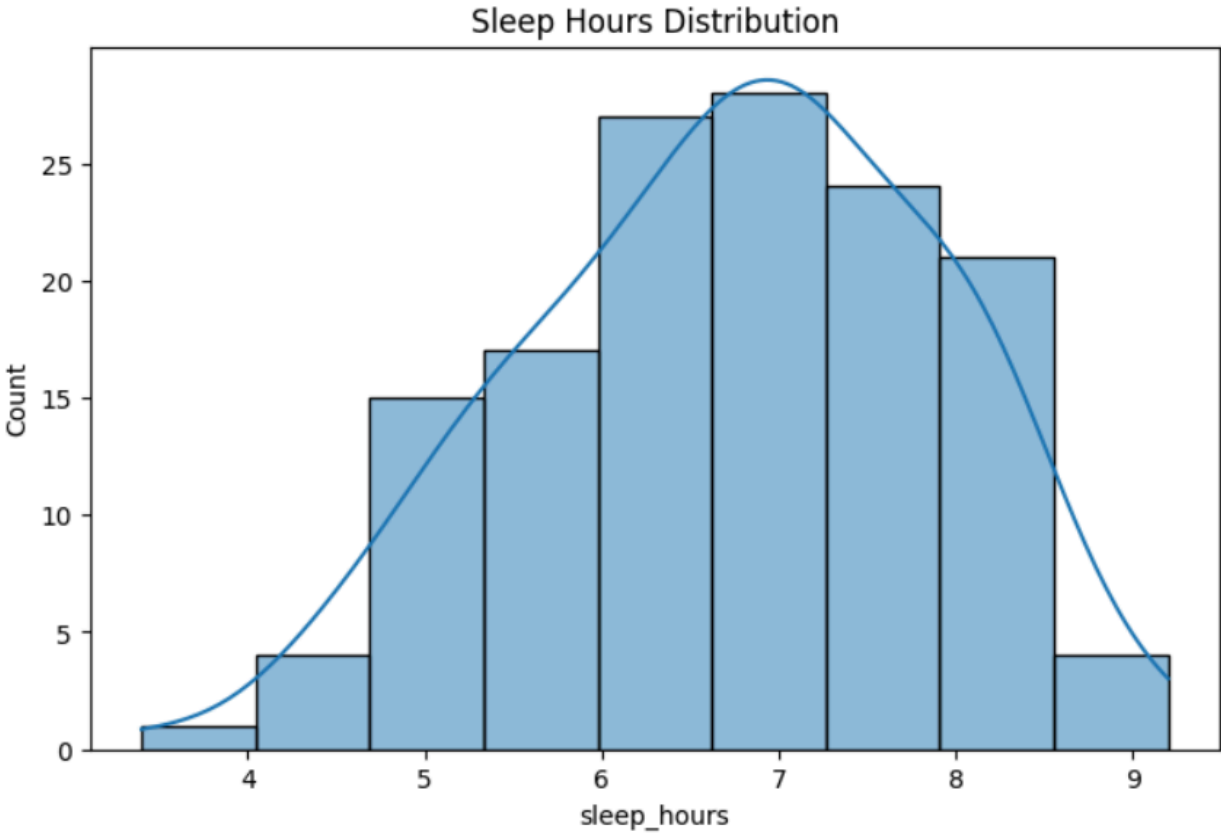


Figure 9 : Sleep Hours Distribution Histogram

6.1.5 Stress Score Distributions

The histogram for stress_score showed a widely distributed range on the normalised scale of 1-10, but there was a small bias in favour of stress_score values from 6 to 9.

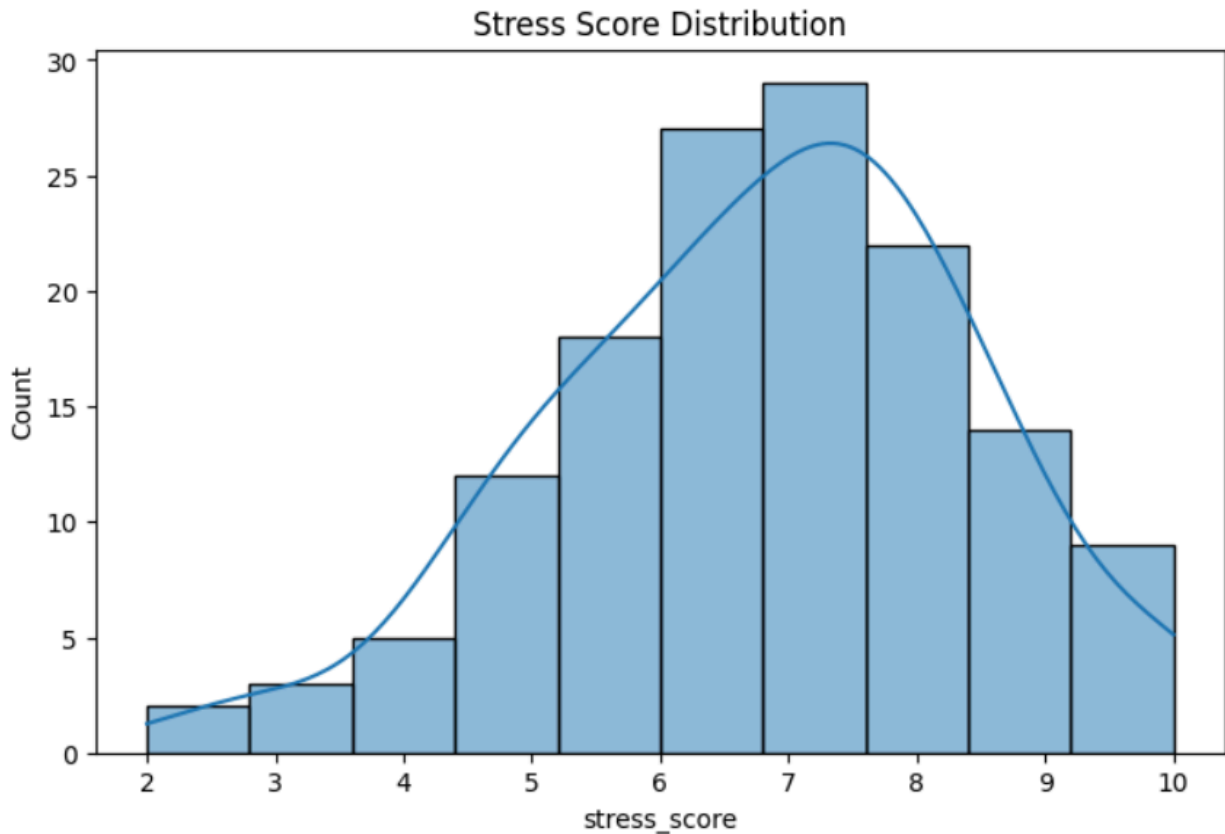


Figure 10 : Stress Score Distribution Histogram

6.1.6 Productivity Drop Distributions

The histogram `productivity_drop_percent` showed a right skewed pattern because many individuals suffered a negligible or moderate level of productivity drop falling within the range of 0-20%.

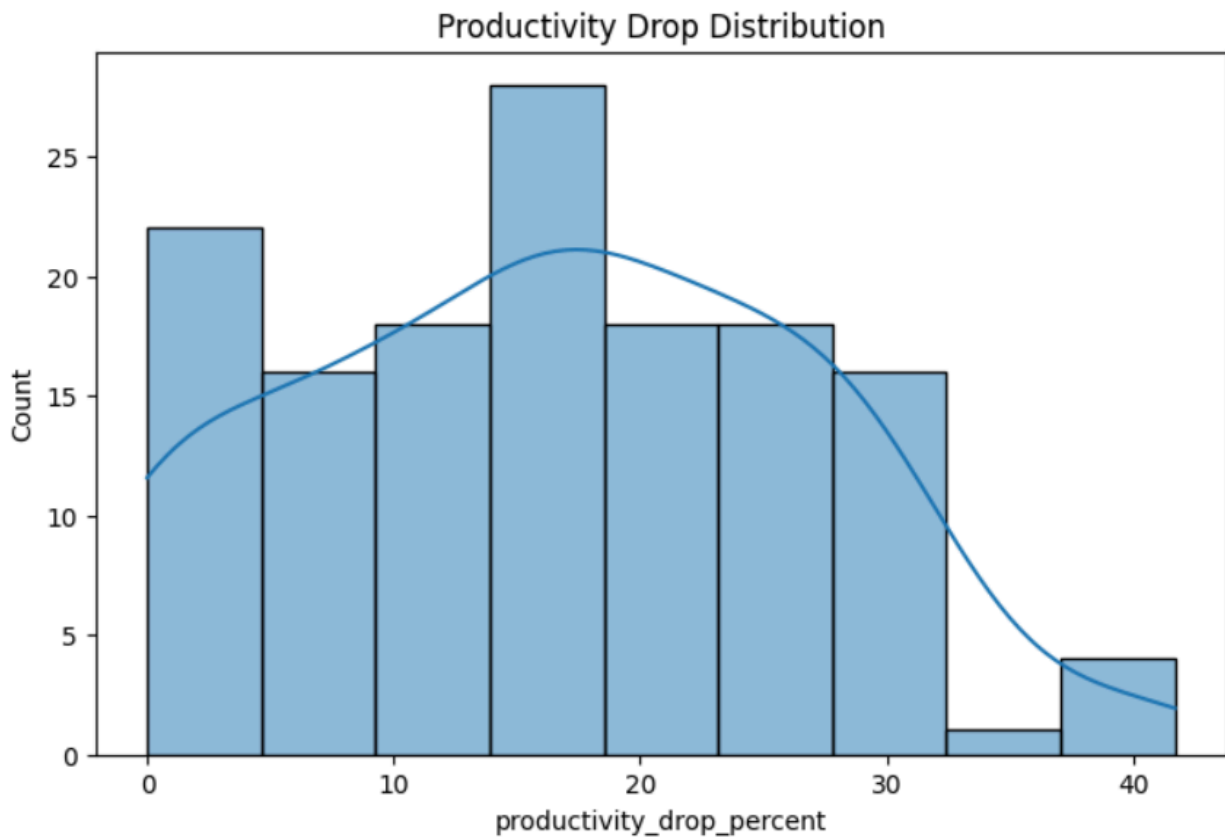


Figure 11 : Productivity Drop Distribution Histogram

6.1.7 Playtime vs Addiction Severity

The `daily_playtime_hours` by class graph depicted through the use of a boxplot was able to offer an easy visualization of the association between the two. The severe class had a high median value for

daily_playtime_hours as well as an interquartile range whereas the mild class had low values for daily_playtime_hours.

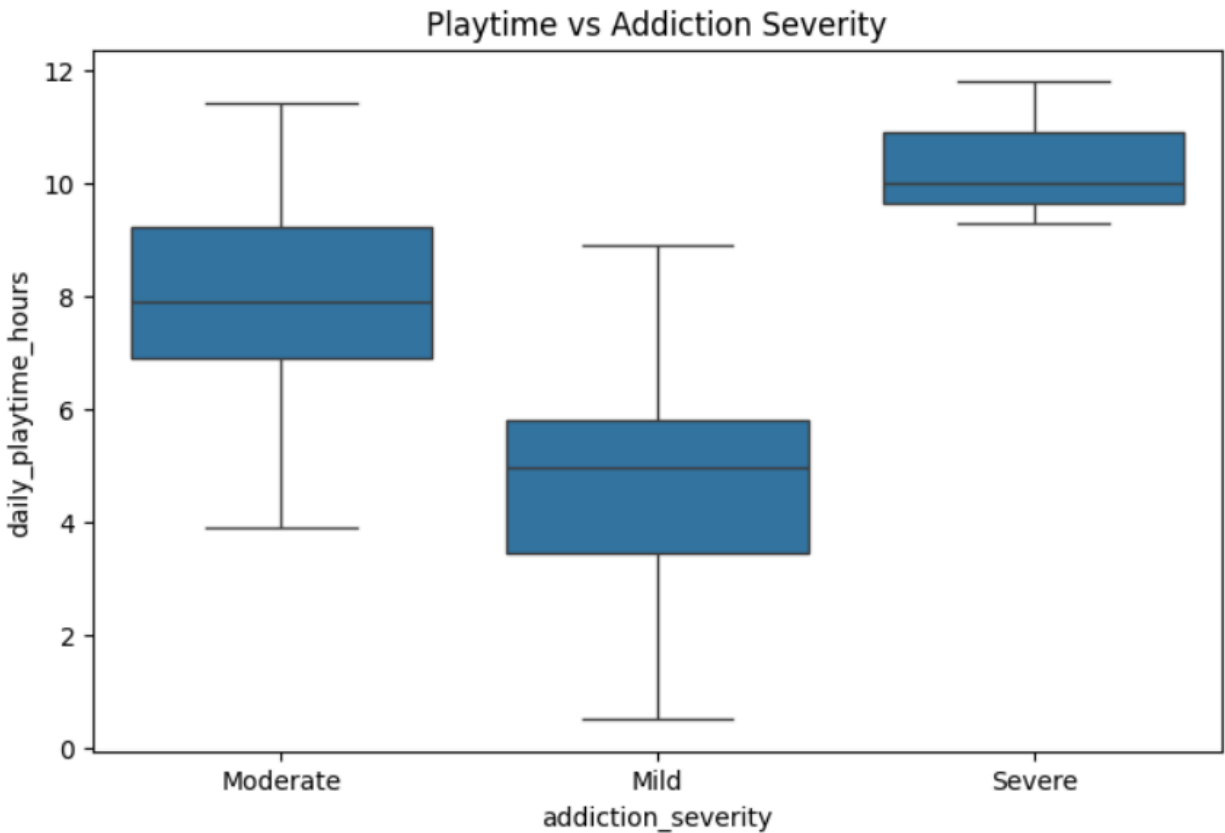


Figure 12 : Boxplot of Daily Playtime Hours by Addiction Severity Class

6.2 Correlation Analysis

The Pearson correlation matrix was done for all numeric variables. The filtering of variables was based on the absolute value of the correlation that was 0.15 against the variable named addiction_severity. The final plot was made using coolwarm color scheme.

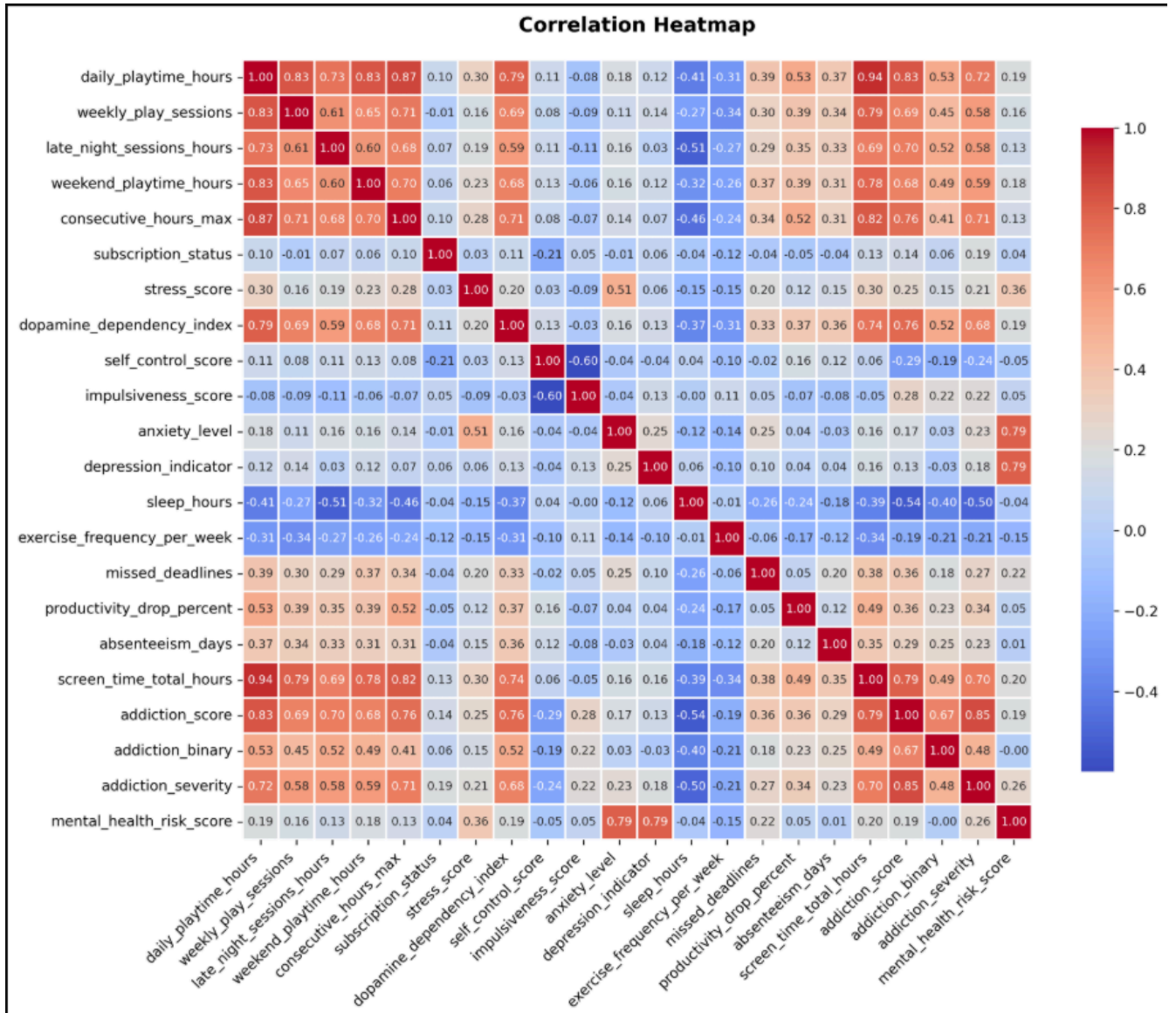


Figure 13 : Correlation Heatmap

6.3 Random Forest Model

6.3.1 Train Test Split

Following data pre processing and exploratory analysis the dataset was prepared for supervised classification. Identifier and redundant columns `user_id`, `addiction_score`, `addiction_binary` were removed from the feature matrix. The remaining 45 features formed the input `X` while `addiction_severity` served as the target variable `y`. After the 80/20 stratified split 112 records were used for training and 29 for testing.

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("Training set size:", X_train.shape[0])
print("Testing set size:", X_test.shape[0])

Training set size: 112
Testing set size: 29
```

Figure 14 : Output confirming 112 training and 29 testing instances

6.3.2 Model Training

Random Forest classifier was trained using 500 trees maximum depth was set to 12 with minimum number of leaves as 2 along with balancing of class weights for tackling class imbalance. Class proportions were maintained during splitting into train and test data sets.

```
model = RandomForestClassifier(  
    n_estimators=500,  
    max_depth=12,  
    min_samples_leaf=2,  
    min_samples_split=5,  
    class_weight="balanced",  
    random_state=42  
)  
  
# Train model  
model.fit(X_train, y_train)  
  
print("Model Training Completed")  
  
print(  
    "Random Forest Trees:",  
    len(model.estimators_)  
)  
  
Model Training Completed  
Random Forest Trees: 500
```

Figure 15 : Output confirming successful model training

7. Analysis of Results and Conclusion

7.1 Model Performance Metrics

Evaluation on the hold out set was done using accuracy, precision, recall, and f1 score on a per class basis as reported by Scikit Learn `classification_report`. The results are presented in the table below.

Metric	Value	Interpretation
Accuracy	89.66%	26 of 29 test cases correctly classified
Precision (Mild)	93%	93% of predicted Mild cases were truly Mild
Recall (Mild)	93%	93% of actual Mild cases correctly identified
F1-Score (Mild)	93%	Strong balanced performance on the Mild class
Precision (Moderate)	87%	87% of predicted Moderate cases were truly Moderate
Recall (Moderate)	93%	93% of actual Moderate cases correctly identified
F1-Score (Moderate)	90%	Strong balanced performance on Moderate class
Precision (Severe)	0%	Severe class not predicted – only 1 test sample, none predicted
Weighted F1-Score	88%	Overall harmonic mean accounting for class imbalance
Class Weighting	Balanced	Applied to compensate for the very small Severe class (3 records)

Table 2 : Random Forest Model Performance Summary

	precision	recall	f1-score	support
0	0.93	0.93	0.93	14
1	0.87	0.93	0.90	14
2	0.00	0.00	0.00	1
accuracy			0.90	29
macro avg	0.60	0.62	0.61	29
weighted avg	0.87	0.90	0.88	29

Figure 16 : Full Classification Report Output per class precision, recall and F1-score

7.2 Confusion Matrix

The confusion matrix reveals the precise distribution among the three classes in the 29 test instances. For Mild 13 classified correctly 1 classified as Moderate. For Moderate 13 classified correctly 1 classified as Mild. And for Severe 0 classified correctly the one Severe test instance classified as Moderate. This was expected since Severe was represented only by 3 training instances so it was very hard to classify reliably.

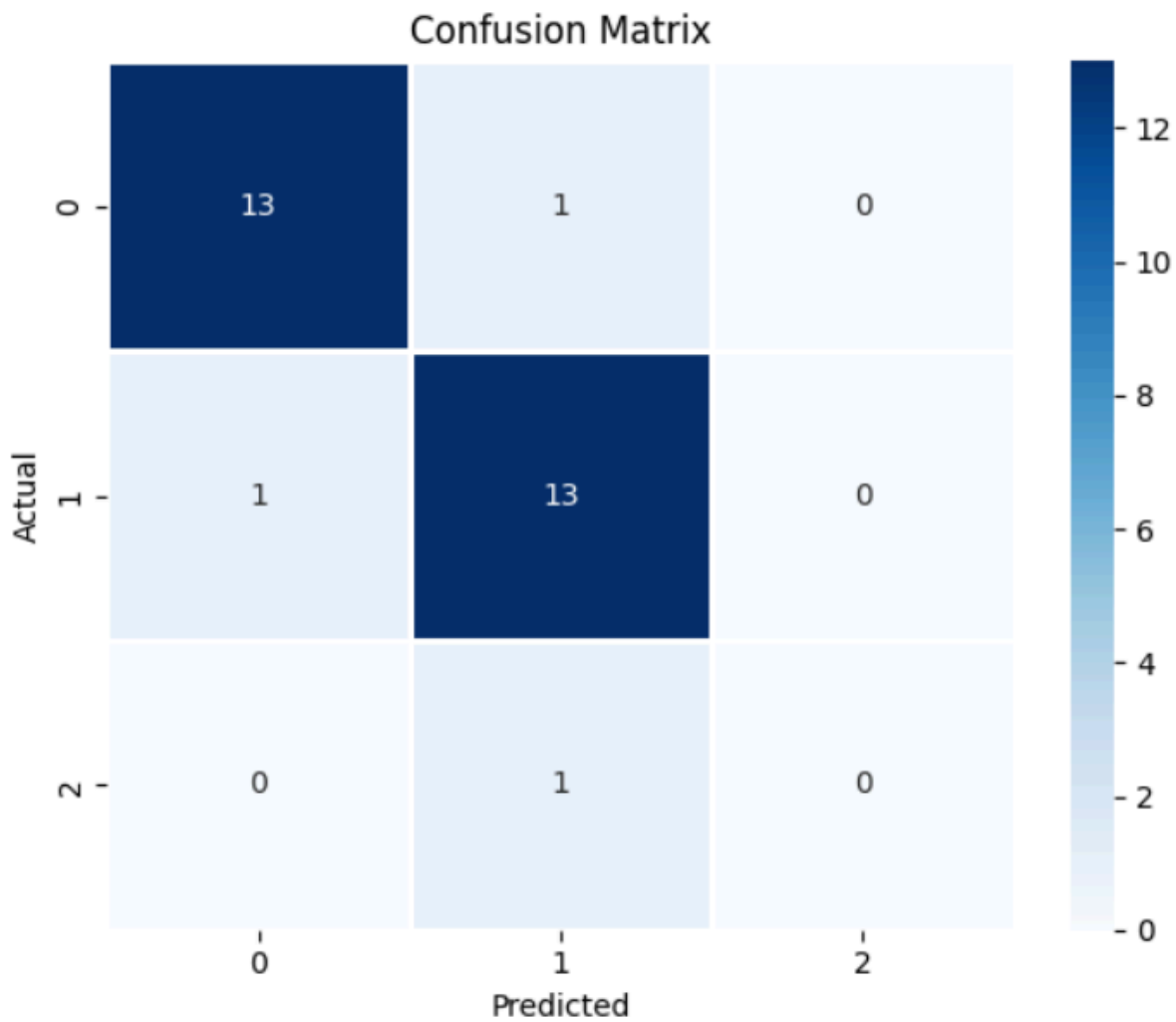


Figure 17 : Confusion Matrix Heatmap – actual vs predicted addiction severity classes

7.3 Feature Importance Analysis

One of the most valuable outputs of the Random Forest model is its built in feature importance measure derived from the mean decrease in impurity across all decision trees. The top 10 features are visualised below.

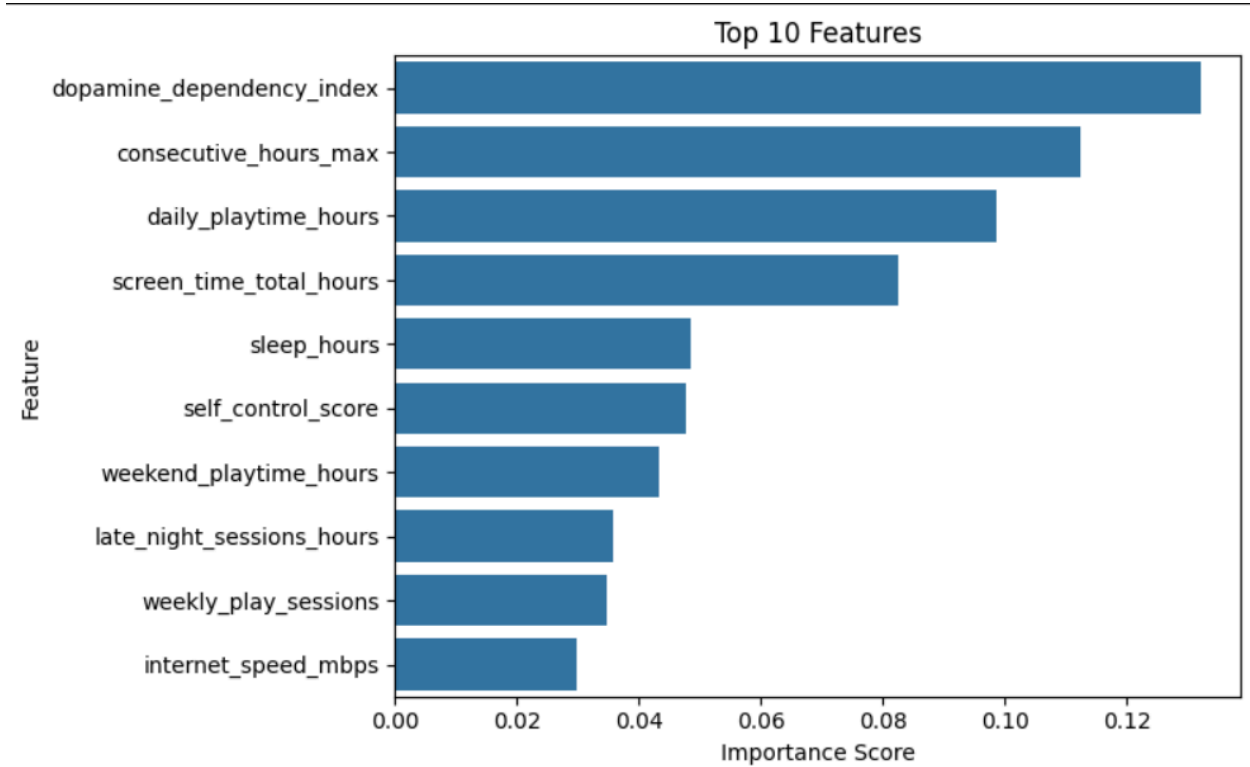


Figure 18 : Top 10 Features Bar Chart

7.4 Prediction Confidence Distribution

Confidence in predictions was measured by obtaining the maximum class probability per predicted instance from `model.predict_proba()`. The mean confidence level for all 29 instances of predictions was calculated at 78.12% .

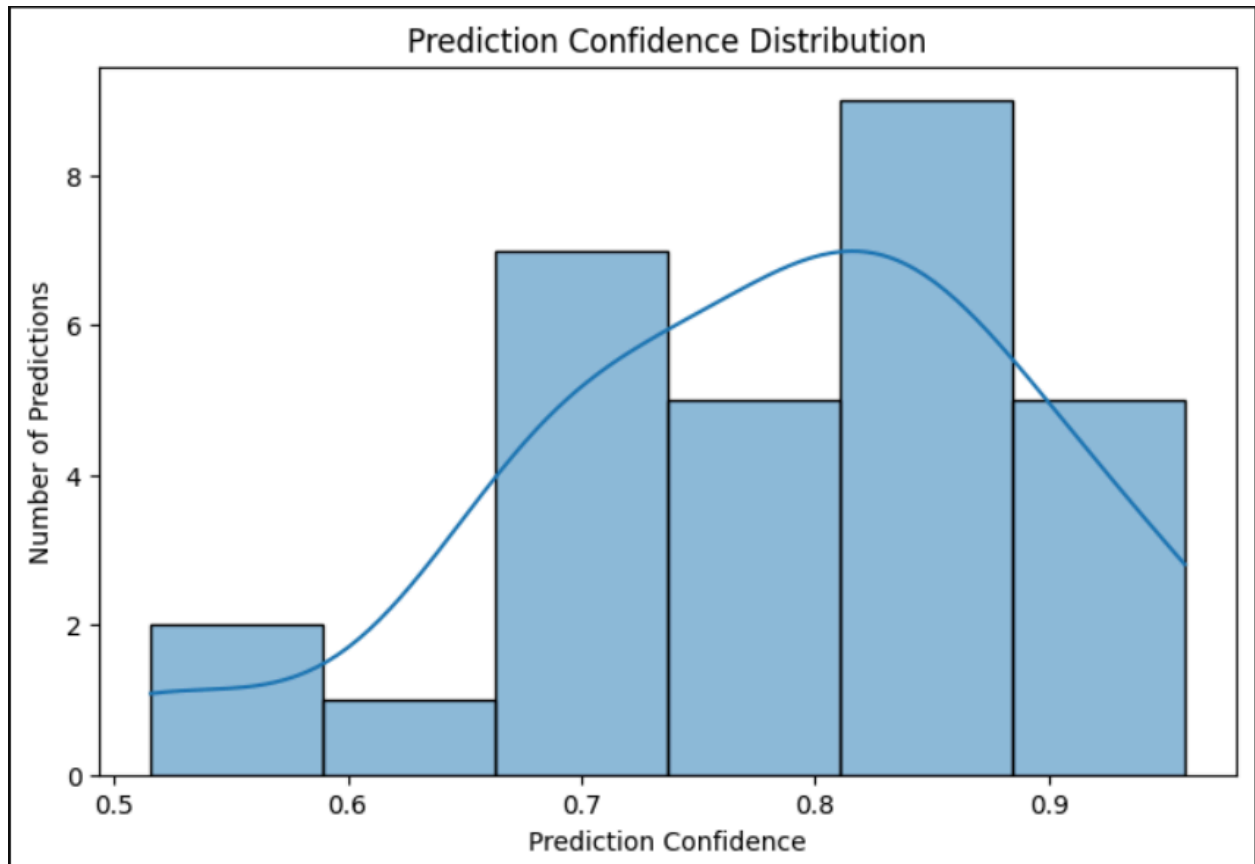


Figure 19 : Prediction Confidence Distribution Histogram

7.5 Error Analysis

The classification errors have been found based on the difference between the predictions and actual values. The Classification errors has been made only in 3 out of 29 samples with the percentage of errors being 10.34%.

Total Misclassified Cases: 3			
Error Rate: 10.34%			
	Actual	Predicted	Probability
14	2	1	0.832
20	1	0	0.521
28	0	1	0.660

Figure 20 : Misclassified Cases Table with Predicted Probability Scores

7.6 Model Insights Summary

Insights on model performance were created using a program that summarizes important indicators into one output block. From the output provided the performance of the model was validated by determining accuracy the most significant attribute confidence in prediction, performance grade, and instances of misclassification.

```
MODEL INSIGHTS SUMMARY
Model Accuracy: 89.66%
Most Influential Feature: dopamine_dependency_index
Feature Importance Score: 0.1320
Average Prediction Confidence: 78.12%
Overall Model Performance: Excellent
Total Features Used: 45
Misclassified Cases: 3
```

Figure 21 : Model Insights Summary Output

7.7 Conclusion

Results from this study indicate that the Random Forest classifier performs well in predicting gaming addiction severity. This is evident since the model had an accuracy rate of 89.66%. Additionally, the classification performed well in the Mild and Moderate categories but not in the Severe category due to only three cases within this category. From feature importance analysis, it was noted that some of the factors which contributed highly to prediction included dopamine_dependency_index, consecutive_hours_max, daily_playtime_hours, screen_time_total_hours, and sleep_hours.

The main issue associated with this study was that there were many missing values, reducing the number of records used in analysis from 250 to 141 after data cleaning. Another issue is that there were very few samples in the Severe category leading to imbalanced data. Thus, future studies should consider acquiring more data samples within the Severe category through the use of SMOTE technique.

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